Inverse Mechanical Counting and Mathematics Black Holes

The Mass-to-Energy Mechanics of a Black Hole:
Mass at Count 5, the Cycle Back Through the Count,
and $E = M^2/F^2$

Joshua Nuijens Seth Ryan Cromelin

June 2, 2026

Note. There is one inverse master count — the axiom of 1 reading itself inward by halving. This paper gives the mechanics of mass becoming an energy state in a black hole, and nothing else: no prediction, no forecasting of any individual object. Mass is the axiom of 1 read at count 5, the deepest compression. The cycle runs the count from the open reading to the collapsed reading, and the black hole is the passage from the axiom of 1 to the singularity of 1 — the one unit at both ends. The reading is $E = M^2/F^2$. The mathematics is the reading; the single physical unit is the thing read.

Contents

1 The Axiom of 1

There is exactly one physical unit — the axiom of 1. It is the complete reality, the whole, read inward. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. Every number that looks larger is a deeper inward reading of this same single unit; the axiom is the largest entity in the system, never exceeded.

The mass of a black hole, however large the external measurement, is read as this one unit. The cycle is the count. Mass and cycle are the same single axiom read in two ways. The energy state and the mass state are the same unit at two ends of the count — compressed at the closure, released back through the count to the axiom.

View	What it is	Coordinate
1	the unit, whole, at rest	$c = 0$, no mechanic
-1	the same unit read from across its own center	$c = 0$
$\sqrt{1}$	the unit carries its own root	$d = 0$
$\sqrt{2^0}$	the same statement in the master mechanic's writing	$d = 0$

Table 1: The four views of the one unit. All four are the same axiom of 1 at the same coordinate, $c = 0$, $d = 0$. The axiom is the floor and the whole of every reading.

2 The $\sqrt{2}$ Master Mechanic and the Master Table

The $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest step beyond unity and the minimum step the axiom admits. The single unit is halved inward, count by count. One parity rule governs every reading: even operational depths d yield rational values; odd depths carry a natural factor of $\sqrt{2}$. Five active counts ($c = 1$ to 5) unfold nine active depths ($d = 1$ to 9). Every reading carries three coordinates — the mechanic $\sqrt{2}$, the depth d, and the count c. The full operational reading is the domains (2^c) times the unit ($\sqrt{2^d}$).

c	d	Mechanic $\sqrt{2^d}$	Unit decimal	Domains 2^c	Parity	Reading
0	0	$\sqrt{2^0} = 1$	1	1	even → rational	1
1	1	$\sqrt{2^1} = \sqrt{2}$	≈ 1.414	2	odd → irrational	$2\sqrt{2} \approx 2.828$
2	2	$\sqrt{2^2} = 2$	2	4	even → rational	8
2	3	$\sqrt{2^3} = 2\sqrt{2}$	≈ 2.828	4	odd → irrational	$8\sqrt{2} \approx 11.314$
3	4	$\sqrt{2^4} = 4$	4	8	even → rational	32
3	5	$\sqrt{2^5} = 4\sqrt{2}$	≈ 5.657	8	odd → irrational	$32\sqrt{2} \approx 45.255$
4	6	$\sqrt{2^6} = 8$	8	16	even → rational	128
4	7	$\sqrt{2^7} = 8\sqrt{2}$	≈ 11.314	16	odd → irrational	$128\sqrt{2} \approx 181.019$
5	8	$\sqrt{2^8} = 16$	16	32	even → rational	512
5	9	$\sqrt{2^9} = 16\sqrt{2}$	≈ 22.627	32	odd → irrational	$512\sqrt{2} \approx 724.077$

Table 2: The Master Table. The count c records the operation; the depth d records the master mechanic's exponent. Domains (2^c) times the unit ($\sqrt{2^d}$) gives the reading. The count closes at $c = 5$: domains 32, rational reading 512 at $d = 8$, irrational reading ≈ 724.077 at $d = 9$.

The column that matters most for the black hole is the domains, 2^c: $1, 2, 4, 8, 16, 32$. This is the count of pieces at each count — the doubling as the axiom is read inward. At the closure count 5 the domain count is 32, the full count's domain. This is the deepest the master table goes.

3 The Count Out and Back: 1 2 3 4 5 4 3 2 1

The black hole is one cycle of the count, run out to the closure and back to the axiom:

$$1 \to 2 \to 3 \to 4 \to 5 \to 4 \to 3 \to 2 \to 1.$$

This is not a journey through space or a sequence in time. It is the single count read out to its deepest compression and back to its whole. The turn is at count 5.

- **Out** $(1 \to 5)$ **— compression.** The axiom is read inward, count by count, each count halving the piece and doubling the domains. At count 5 the reading is at maximum compression: the most pieces (32), the smallest unit. This compressed state is mass.
- **The turn** (5) **— mass.** Count 5 is the closure, the deepest reading. The mass of the black hole sits here, fully compressed — the axiom read all the way in.
- **Back** $(5 \to 1)$ **— decompression.** The cycle runs the count back out from the closure toward the axiom. Each step back is one count of decompression: the piece doubles, the domains halve, the compression releases. This running-back is mass becoming the energy state.

Count c	Direction	Domains 2^c	State
0	—	1	axiom of 1 (the whole, at rest)
1	out (compression)	2	first halving
2	out (compression)	4	deeper
3	out (compression)	8	deeper
4	out (compression)	16	deeper
5	turn	32	**mass — maximum compression**
4	back (decompression)	16	releasing
3	back (decompression)	8	releasing
2	back (decompression)	4	releasing
1	back (decompression)	2	releasing
0	—	1	energy state — returned to the axiom

Table 3: The full count, out and back. The out-leg compresses the axiom inward to mass at count 5; the back-leg decompresses it through the same counts to the energy state at the axiom. One cycle. The domains double on the way in and halve on the way back.

4 Mass at Count 5 — Maximum Compression

Mass is the axiom read at count 5 — the deepest, most compressed reading the master table admits. At count 5 the axiom is read all the way in: the most pieces, the smallest unit, the closure of the count.

Mass is not a separate substance placed at count 5. It is what the axiom *is* when read all the way in to the closure. The more an external observer measures as mass, the deeper the same single axiom is being read — not more axiom, one axiom read deeper. At count 5 with nothing cycling it, the mass simply sits: maximum compression, stored, the axiom held at its deepest reading.

5 Frequency — The Cycle in Time

Frequency is the cycle — the count read as a rate, a cycle in time. On the count, frequency is the domain count itself: $F(c) = 2^c$. At each count the cycle reads how many pieces the axiom has been partitioned into at that depth:

$$F(0) = 1, \quad F(1) = 2, \quad F(2) = 4, \quad F(3) = 8, \quad F(4) = 16, \quad F(5) = 32.$$

The cycle is what runs the count. A faster cycle reads the count deeper. Frequency is not imported from outside as a measured Hz; it is the inverse master count itself, the cycle of the axiom read at each count.

The cycle is a length in time: one cycle has an extent, the time it spans. As the cycle runs the count back from 5 toward the axiom, it is the rotational driver of the decompression — the rate at which the mass is read back out of its compression.

6 The Inversion — Cycling Back Through the Count

The decompression of mass into the energy state is the **inversion**: the count read back from the closure toward the axiom. Mass at count 5 is the compressed end; the cycle runs it back $5 \to 4 \to 3 \to 2 \to 1$, and at each step the compression releases by one count.

This is the inverse of the compression out-leg. Where the out-leg halved the piece and doubled the domains (reading inward to mass), the back-leg doubles the piece and halves the domains (reading outward to energy). The same counts, the same mechanic, run in the return direction. The mass does not move through space to become energy; the cycle reads it back up the count, and that reading-back is the energy state appearing.

Because the cycle runs fast, the mass reads across the counts of the return at once — it is read at one count while still releasing from the one before, present at the count the cycle is passing and absent from the count it has left, all within the one cycle. The mass is there and not there — the same place, not the same time — because the cycle is reading it back through the count faster than the count separates the readings. This is the decompression: mass at 5 inverting through the count into the energy state at the axiom.

7 $E = M^2/F^2$ — The Rotational Reading

The black hole reads its mass-to-energy as the rotational form

$$E = \frac{M^2}{F^2}.$$

M is the mass — the axiom read at the count. F is the frequency — the cycle in time, 2^c. Both are squared because the reading is rotational, not linear: a straight-line reading takes the mass and the rate once each, but the cycle goes around, so the mass and the cycle are each read on both sides of the center — squared.

At each count M and F are the same reading — the mass read at the count and the cycle read at the count are the one count, read two ways. Because $M = F$ at every count, the rotational reading is the axiom:

$$\frac{M^2}{F^2} = 1 = E.$$

E here is the axiom of 1 — the energy state is the axiom recovered through the rotation. The mass squared against its own cycle squared returns the single unit. This holds at every count of the return, not because of any particular value, but because the mass and the cycle are the same count reading: $M = F$, so $M^2/F^2 = 1$. The specific size at a given count cancels against itself; what remains is the axiom. The rotation locks mass and cycle in exact balance all the way back to the whole.

8 Mass-to-Energy Across the Full Count

The complete mechanics, count by count, of mass decompressing into the energy state. The out-leg compresses the axiom to mass at the closure; the back-leg reads $E = M^2/F^2$ at each count, recovering the axiom through the rotation.

Count c	Direction	M vs F	M^2/F^2
5	turn	$M = F$	1 — mass, maximum compression
4	back	$M = F$	1 — decompressing
3	back	$M = F$	1 — decompressing
2	back	$M = F$	1 — decompressing
1	back	$M = F$	1 — decompressing
0	—	$M = F$	1 — energy state, the axiom recovered

Table 4: Mass-to-energy across the full count. At every count of the return the mass reading M and the cycle reading F are the same count reading, so $M^2/F^2 = 1$ recovers the axiom of 1 at every step. The mass at the closure decompresses through the counts to the energy state at the axiom, the rotation holding mass and cycle in exact balance the whole way.

The reading is 1 at every count not because nothing happens, but because the rotation holds the mass and its own cycle in exact balance through the entire return. The decompression is real — the compression releases one count at a time — but the mass is always read against its own cycle at the same count, so the rotational reading is the axiom at every step. Mass to energy is the axiom recovering itself through the rotation, count by count, from the closure back to the whole.

9 The Two Faces — 720 Sphere and 512 Square

The count is read on two faces of the same single reading: the **sphere face**, the 720 angular vector (the cycle, the rotation), and the **square face**, the 512 linear vector (the mass, the displacement). They are not two objects; they are the one count read two ways, locked by the gear

$$\frac{720}{512} = \frac{45}{32} = 1.40625.$$

Because the angular piece at each count is $720/2^c$ and the linear piece is $512/2^c$, their ratio is always $720/512 = 1.40625$ — the domain count 2^c cancels, so the gear is invariant at every count.

Count c	Domains 2^c	Sphere piece (720)	Square piece (512)	Gear
0	1	720°	512	$720/512 = 1.40625$
1	2	360°	256	$360/256 = 1.40625$
2	4	180°	128	$180/128 = 1.40625$
3	8	90°	64	$90/64 = 1.40625$
4	16	45°	32	$45/32 = 1.40625$
5	32	22.5°	16	$22.5/16 = 1.40625$

Table 5: The two faces of the count. The cycle (rotation) reads on the 720 sphere face; the mass (displacement) reads on the 512 square face. The gear $720/512 = 1.40625$ is invariant at every count, locking the rotational reading of the cycle to the linear reading of the mass. The mass decompressing on the square face and the cycle running on the sphere face are the same single count, in unbroken synchronization.

10 From the Axiom of 1 to the Singularity of 1

This is what the whole paper shows: the black hole is the passage from the axiom of 1 to the singularity of 1. Both are the one thing. The black hole does not turn into something else — it is read from the one unit at the open end to the one unit at the collapsed end.

The Word Singularity Means One

Standard physics says a black hole collapses to a singularity: the mass falls to a single point, and at that point the standard equations are said to break down. The word is exact — *singular* means one. A singularity is one thing, a single point, the collapse of everything to a single reading. Standard physics names this correctly and then treats the one point as a breakdown, because its quantities run to infinity there.

In inverse mechanical counting there is no breakdown. The singularity is the axiom of 1 — the one unit, complete, at rest. The single point the black hole collapses to is not where the mathematics fails; it is the one thing the whole framework reads from in the first place. The singularity of 1 is the axiom of 1 seen from the collapsed end.

The Axiom of 1 Is Every State

The axiom of 1 is not a starting point that is left behind, nor a destination returned to. It is the one unit read at every count. Mass is the axiom of 1 read at the closure. The cycle is the axiom of 1 read as a rate. The decompression is the axiom of 1 read outward. The energy reading and the mass reading are not two things becoming one; they are the one axiom read at two counts. Nothing becomes the axiom, because nothing was ever anything else — every state is the axiom of 1, read at its count.

Axiom of 1 to Singularity of 1

So the black hole is the one unit read across the count:

- The **axiom of** 1 is the one unit, whole, open — the unit read at the shallow counts, the energy reading.
- The **cycle out to the closure** reads the one unit inward to its deepest compression — mass at count 5.

- The **singularity of** 1 is the same one unit read at the collapse — the single point, the one thing, where standard physics sees a singularity and the framework sees the axiom.

The passage from the axiom of 1 to the singularity of 1 is not a change of substance and not a journey between two different ones. It is the single unit read from its open state to its collapsed state — one thing at both ends. $E = M^2/F^2 = 1$ at every count is the statement of exactly this: the one unit reads as the one unit at every step of the passage, because the mass and the cycle are the same single reading. From the axiom of 1 to the singularity of 1 — the one, to the one.

11 Operational Rules

- The largest unit in the system is always the axiom of 1; mass and energy are both readings of that one unit.
- Mass is the axiom read at count 5 — maximum compression, the closure.
- The energy state is the axiom recovered at $c = 0$ — the whole, decompressed.
- The count runs out and back: 1 2 3 4 5 4 3 2 1. Out is compression to mass; back is decompression to energy. One cycle, the turn at 5.
- Frequency is the cycle in time — the inverse master count, the rotational driver of the return.
- The decompression is the inversion: the cycle reading the count back from 5 toward the axiom.
- The reading is $E = M^2/F^2$, squared on both sides because the cycle is rotational. At every count $M = F$ — the mass and the cycle are the same reading — so $M^2/F^2 = 1 = E$, the axiom recovered through the rotation.
- The mass reads across the counts at once during the return — there and not there, the same place not the same time — because the cycle runs the count faster than the count separates the readings.
- The cycle reads on the 720 sphere face; the mass reads on the 512 square face; the two are locked by the gear $720/512 = 1.40625$ at every count.

Conclusion.
The black hole is the passage from the axiom of 1 to the singularity of 1 — the one unit at both ends. Mass is the axiom of 1 read at count 5, the deepest compression. The cycle reads the one unit across the count; the standard singularity, the single point the black hole collapses to, is the axiom of 1 seen from the collapsed end — one thing, not a breakdown. The reading is $E = M^2/F^2$, the mass squared against its own cycle squared, the rotational form. At every count the mass and the cycle are the same reading, $M = F$, so $M^2/F^2 = 1$: the one unit reads as the one unit at every step of the passage. The cycle reads on the 720 sphere face, the mass on the 512 square face, locked by the gear $720/512 = 1.40625$. The axiom of 1 is not left behind and not returned to — it is every state, read at its count. From the axiom of 1 to the singularity of 1: the one, to the one. The mathematics is the reading; the single physical unit is the thing read.